

$$\begin{aligned}
\tilde{E}(\check{\ell})|REF\rangle_{\pm} &= \sum_{r,v} \left[\xi_{r,1,v} N_{r,1,v,+} (1 - N_{r,1,v,-}) + (1 - \xi_{r,1,v}) \chi_{r,1,v,+}^{\dagger} \chi_{r,1,v,-} - 1 \right] |\check{\ell}\rangle\rangle \\
&\quad + \sum_{r,v} \left[(1 - \xi_{r,2,v}) (1 - N_{r,2,v,+}) N_{r,2,v,-} - \xi_{r,2,v} \chi_{r,2,v,+} \chi_{r,2,v,-}^{\dagger} - 1 \right] |\check{\ell}\rangle\rangle \\
&= \sum_{r,v} \left[\xi_{r,1,v} N_{r,1,v,+} \otimes 1_- + (1 - \xi_{r,1,v}) \chi_{r,1,v,+}^{\dagger} \chi_{r,1,v,-} + (1 - \xi_{r,2,v}) 1_+ \otimes N_{r,2,v,-} - \xi_{r,2,v} \chi_{r,2,v,+} \chi_{r,2,v,-}^{\dagger} \right] \\
&\quad - \sum_{r,v} \left[\xi_{r,1,v} N_{r,1,v,+} N_{r,1,v,-} + (1 - \xi_{r,2,v}) N_{r,2,v,+} N_{r,2,v,-} \right] \text{ mod constant}
\end{aligned}$$

Maybe we should consider the covariance matrix evolution?

$$\begin{array}{c} t'' \\ \text{---} \textcircled{H} \text{---} t' \end{array} \quad \begin{array}{c} t \\ \text{---} \textcircled{G} \text{---} b \end{array} \longrightarrow \begin{array}{c} t'' \\ \text{---} \textcircled{G'} \text{---} b \end{array}$$

$$G' = \begin{pmatrix} H_{t''t'} \\ G_{bb} \end{pmatrix} - \begin{pmatrix} H_{t''t'} \\ G_{bt} \end{pmatrix} \begin{pmatrix} H_{t't'} - 1 \\ -1 \quad G_{tt} \end{pmatrix}^{-1} \begin{pmatrix} H_{t''t'} \\ G_{bt} \end{pmatrix}^{\dagger}$$

$$\begin{aligned}
[G']_{t''p} &= H_{t''t''} - H_{t''t'} (H_{t't'} - G_{tt}^{-1})^{-1} H_{t't''} \\
&= H_{22} - H_{21} (H_{11} - G_{tt}^{-1})^{-1} H_{12}
\end{aligned}
\quad \begin{aligned}
&(H_{11} - G_{tt}^{-1}) \\
&- (H_{11} G_{tt} - 1) G_{tt}^{-1}
\end{aligned}$$

$$\text{Suppose } H \text{ has the form } H = \begin{pmatrix} H_{11} & H_{12} \\ H_{21} & H_{22} \end{pmatrix} = \begin{cases} \begin{pmatrix} -P & 1-P \\ 1-P & P \end{pmatrix} & (\text{occupied}) \\ \begin{pmatrix} P & 1-P \\ 1-P & -P \end{pmatrix} & (\text{unoccupied}) \end{cases}$$

Where $P = \chi \chi^{\dagger}$ is a Rank-1 projector, then we can apply Sherman-morrison

Because H_{tt} is a rank-1 projector, we have

$$\Rightarrow G'_{\pm} = \pm P - (1-P)(\mp P - G^{-1})(1-P),$$

$$\text{Sherman-morrison: } (\mp P - G^{-1})^{-1} = -G \mp \frac{GPG}{1 \pm \text{Tr}(GP)}$$

$$\Rightarrow G'_{\pm} = \pm P + (1-P)G(1-P) \pm \frac{(1-P)GP(1-P)}{1 \pm \text{Tr}(GP)}$$

Suppose $G'_{\pm} = G^* + \delta G'_{\pm}$, $G = G^* + \delta G$

$$\Rightarrow G^* + \delta G'_{\pm} = \pm P + (1-P)G^*(1-P) + (1-P)\delta G(1-P)$$

$$\pm (1-P)(G^* + \delta G)P(G^* + \delta G)(1-P)(1 \pm \text{Tr}(G^*P) \pm \text{Tr}(\delta GP))^{-1}$$

$$= \pm P + (1-P)G^*(1-P) + (1-P)\delta G(1-P)$$

$$\pm (1-P)(G^* + \delta G)P(G^* + \delta G)(1-P)(1 \pm \text{Tr}(G^*P))^{-1} \left(1 \mp \frac{\text{Tr}(\delta GP)}{1 \pm \text{Tr}(G^*P)}\right)$$

$$= \pm P + (1-P)G^*(1-P) + (1-P)\delta G(1-P)$$

$$\pm \left[(1-P)\delta GP G^*(1-P) + (1-P)G^*P\delta G(1-P) \right] \frac{1}{1 \pm \text{Tr}(G^*P)}$$

$$- (1-P)G^*P G^*(1-P) \frac{\text{Tr}(\delta GP)}{(1 \pm \text{Tr}(G^*P))^2} + O(\delta G^2)$$

Collect terms of order δG^0 , δG

$O(\delta G^0)$: $G^* = \pm P + (1-P)G^*(1-P)$ (*)

$O(\delta G)$: $\delta G' = (1-P)\delta G(1-P) \pm \left[(1-P)\delta GP G^*(1-P) + (1-P)G^*P\delta G(1-P) \right] \frac{1}{1 \pm \text{Tr}(G^*P)}$
 $- (1-P)G^*P G^*(1-P) \frac{\text{Tr}(\delta GP)}{(1 \pm \text{Tr}(G^*P))^2}$ (**)

Consider (*)

$$G^* = \pm P + (1-P)G^*(1-P) \Rightarrow \overset{\text{left multiply}}{(1-P)G^*} = \overset{\text{right multiply}}{(1-P)G^*(1-P)} = G^*(1-P)$$

$$\Rightarrow [G^*, P] = 0$$
 (***)

Apply (***) to (**)

$$\delta G' = (1-P)\delta G(1-P) \pm \overbrace{\left[(1-P)\delta GP G^*(1-P) + (1-P)G^*P\delta G(1-P) \right]}^{=0} \frac{1}{1 \pm \text{Tr}(G^*P)}$$

$$- \underbrace{(1-P)G^*P G^*(1-P)}_{=0} \frac{\text{Tr}(\delta GP)}{(1 \pm \text{Tr}(G^*P))^2}$$
 (**)

$$\Rightarrow \underline{\delta G' = (1-P) \delta G (1-P)} \text{ (****)}$$

Analyzing the particle 'pump in/out' map leads to an identical result
i.e.

$$G'_\pm = (1-P)G(1-P) \pm P, \text{ with } +/- \leftrightarrow \text{occ/unocc}$$

A perturbative expansion about a fixed point $G = G^* + \delta G$ leads to (****)